# Module 8: SLAAC and DHCPv6

Switching, Routing and Wireless Essentials v7.0 (SRWE)

# Module Objectives

**Module Title:** SLAAC and DHCPv6

**Module Objective:** Configure dynamic address allocation in IPv6 networks.

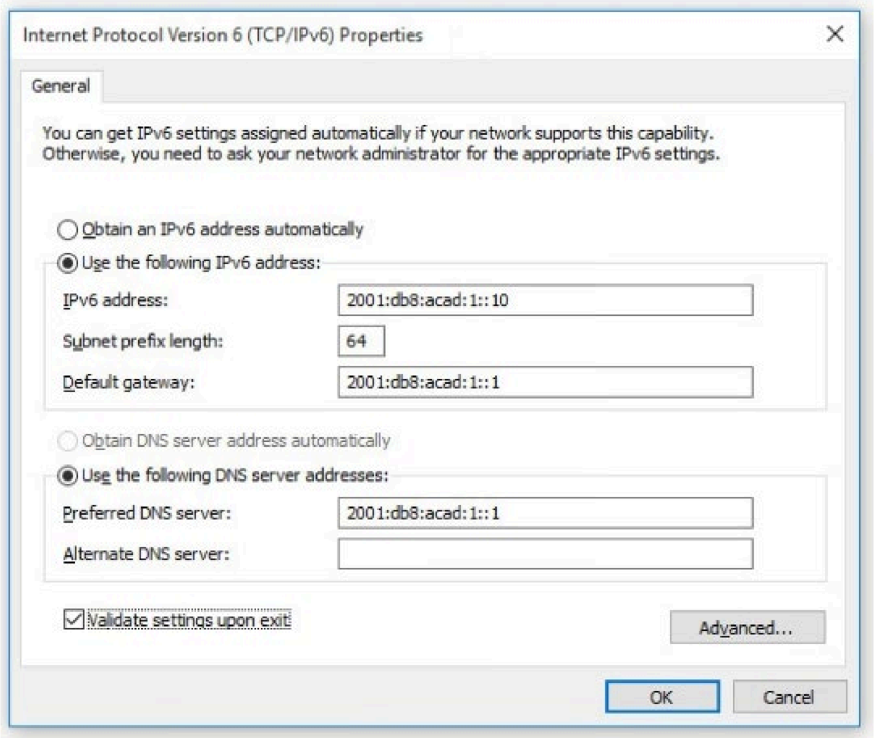
| Topic Title | Topic Objective |
| --- | --- |
| **IPv6 Global Unicast Address Assignment** | Explain how an IPv6 host can acquire its IPv6 configuration. |
| **SLAAC** | Explain the operation of SLAAC. |
| **DHCPv6** | Explain the operation of DHCPv6 |
| **Configure DHCPv6 Server** | Configure a stateful and stateless DHCPv6 server. |

# 8.1 IPv6 GUA Assignment

# IPv6 Host Configuration

On a router, an IPv6 global unicast address (GUA) is manually configured using the **ipv6 address** *ipv6-address/prefix-length* interface configuration command.

- A Windows host can also be manually configured with an IPv6 GUA address configuration, as shown in the figure.
- However, manually entering an IPv6 GUA can be time consuming and somewhat error prone.
- Therefore, most Windows host are enabled to dynamically acquire an IPv6 GUA configuration.



# IPv6 Host Link-Local Address

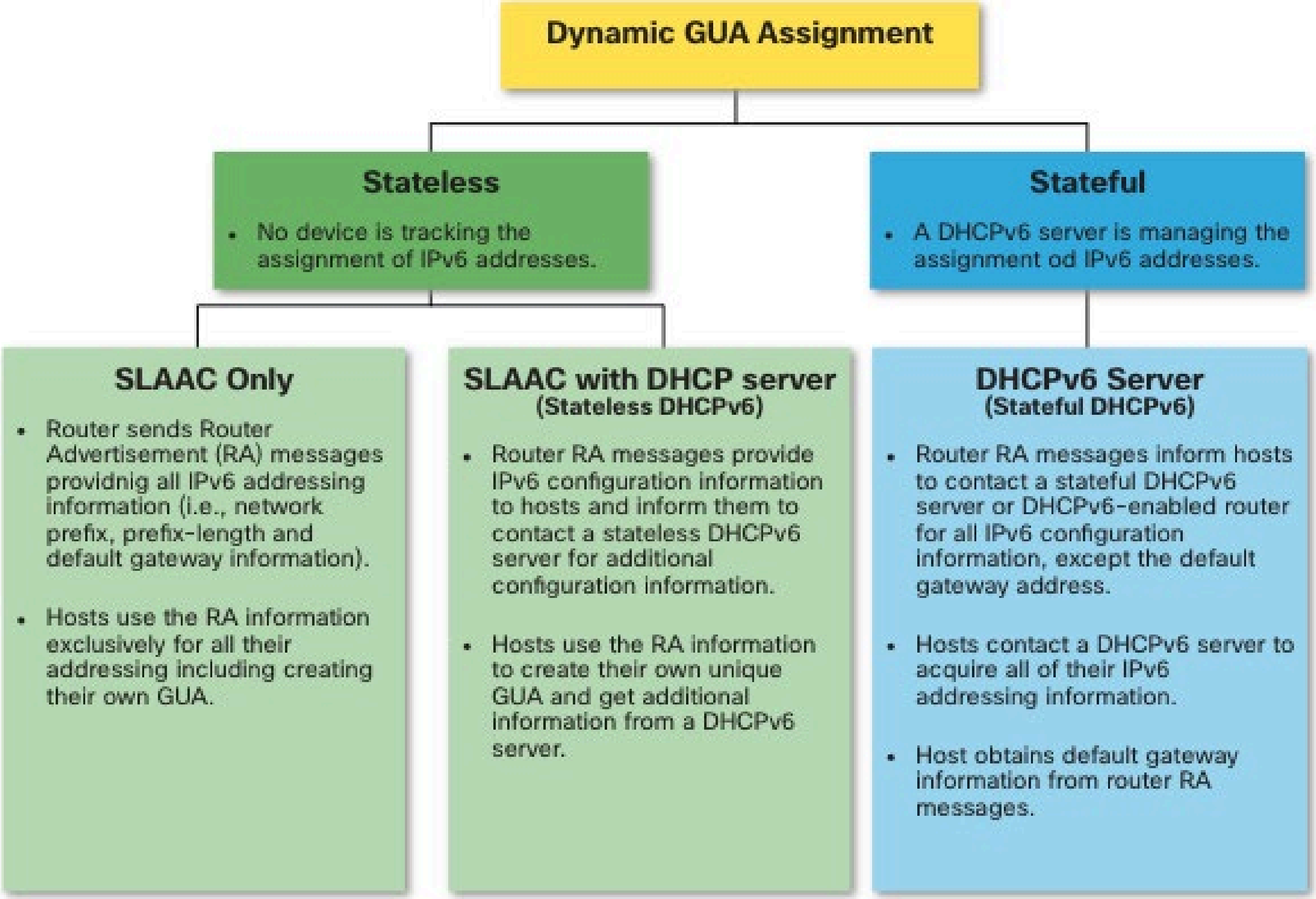
If automatic IPv6 addressing is selected, the host will use an Internet Control Message Protocol version 6 (ICMPv6) Router Advertisement (RA) message to help it autoconfigure an IPv6 configuration.

```
C:\PC1> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet0:
   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . :
   Link-local IPv6 Address . . . . . : fe80::fb:1d54:839f:f595%21
   IPv4 Address. . . . . . . . . . . : 169.254.202.140
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :
C:\PC1>
```

# IPv6 GUA Assignment

- By default, an IPv6-enabled router periodically send ICMPv6 RAs which simplifies how a host can dynamically create or acquire its IPv6 configuration.
- A host can dynamically be assigned a GUA using stateless and stateful services.
- All stateless and stateful methods in this module use ICMPv6 RA messages to suggest to the host how to create or acquire its IPv6 configuration.
- Although host operating systems follow the suggestion of the RA, the actual decision is ultimately up to the host

# IPv6 GUA Assignment



IPv6 GUA Assignment

# Three RA Message Flags

How a client obtains an IPv6 GUA depends on settings in the RA message.

An ICMPv6 RA message includes the following three flags:

- **A flag** - The Address Autoconfiguration
- **O flag** - The Other Configuration flag
- **M flag** - The Managed Address Configuration flag